Samarth Kumar Samal

Data Science | Healthcare Analytics | Software Engineering
Minneapolis, Minnesota | (763) 568-9934 | samal026@umn.edu | linkedin.com/in/samarth-kumar-samal
github.com/samarthkumarsamal | samarthkumarsamal.github.io

PROFESSIONAL SUMMARY

Master of Science in Data Science student at the University of Minnesota Twin Cities with two years of software engineering experience at Dell Technologies. Graduate research includes IMU signal analysis for Parkinson's disease and ICU patient clustering using electronic health record data. Background in data analysis, healthcare research, and software engineering.

EDUCATION

University of Minnesota, Twin Cities — **Expected Spring 2027**
Master of Science in Data Science; Minor in Computer Science; GPA: 3.61/4.00 — *Minneapolis, Minnesota*

Odisha University of Technology and Research — **2018–2022**
Bachelor of Technology in Electrical Engineering; GPA: 3.83/4.00 — *Bhubaneswar, India*

TECHNICAL SKILLS

Programming and Database: Python, R, C++, SQL Server
Data Science Libraries: NumPy, Pandas, Scikit-learn, SciPy
Data Visualization: Matplotlib, Seaborn, Plotly
Healthcare and Signal Analysis: Health Data Analysis, IMU Signals
Cloud and Development Tools: Google Cloud, Google Tools, Docker, Git, GitHub
Development Environments: JupyterLab, Kaggle Notebook, Google Colab, VS Code
Operating Systems: Linux, Windows
Professional Skills: Team Management, Time Management, Attention to Detail, Problem Solving, Hardworking, Research Skills

RESEARCH EXPERIENCE

Graduate Researcher — **May 2026–Present**
Neuromodulation Center, Department of Neurology — *Minneapolis, Minnesota*
Principal Investigator: Prof. Biswaranjan Mohanty

- Analyze and preprocess raw six-axis inertial measurement unit (IMU) data, including three-axis accelerometer and three-axis gyroscope signals, using Python-based signal analysis techniques.
- Develop a reproducible workflow for continuous, unlabeled IMU data with the objective of identifying and labeling movement segments as resting, walking, or turning.
- Perform time-domain and frequency-domain analysis, including power spectral density and spectrogram analysis, to examine signal patterns across movement activities.
- Investigate signal characteristics and extracted features that can support movement classification and later Parkinsonian gait analysis.

Capstone Researcher — **May 2026–Present**
Department of Surgery, Division of Computational Health Sciences — *Minneapolis, Minnesota*
Principal Investigator: Prof. Jocelyn Chi

- Build a reproducible data-processing pipeline using the eICU Collaborative Research Database to combine demographic, laboratory, vital-sign, and illness-severity data for ICU patient analysis.
- Prepare first-24-hour ICU clinical features through cohort selection, data-quality checks, missing-value handling, repeated-measurement aggregation, feature engineering, and scaling.
- Evaluate K-Means clustering solutions using within-cluster sum of squares, silhouette score, cluster size, stability, and clinical interpretability to identify meaningful ICU patient subtypes.
- Characterize patient groups using clinical features, APACHE IVa severity, ICU and hospital mortality, and length of stay to assess the clinical relevance of discovered subtypes.

PROFESSIONAL EXPERIENCE

Software Engineer I — **August 2022–July 2024**
Dell Technologies — *Bengaluru, India*

- Worked in the Dell Update Package (DUP) and USBNIC teams, supporting 5 component areas: iDRAC, USBNIC, PSU, CPLD, and device drivers across Windows, WinPE, and Linux environments.
- Resolved more than 50 Jira issues and contributed to over 100 technical investigations through initial analysis, log review, source-code debugging, issue escalation, testing, and fix validation.
- Contributed to the DUP Modernization initiative by supporting development, testing, and validation of Dell Update Package workflows and component updates.
- Collaborated with the U.S.-based engineering team on the OSM module to analyze issues, validate functionality, and coordinate technical requirements across development and testing teams.
- Set up and supported more than 50 virtual machines and 10 physical servers with the Cloud Server team to meet development, testing, operating-system, and platform requirements.

TECHNICAL PROJECTS

Temporal Clinical Reasoning for Medical Coding — **January 2026**
GitHub Repository | **Live Application**
BioClinicalBERT, Longformer, BiLSTM, MIMIC-III

- Developed BioClinicalBERT, Longformer, and temporal BERT + bidirectional LSTM models for joint multi-label prediction of ICD and CPT medical codes from longitudinal electronic health record clinical notes.
- Constructed admission-level temporal sequences from MIMIC-III by integrating heterogeneous clinical notes, sorting events chronologically, mapping ICD-9 codes to ICD-10, extracting CPT codes, and preparing train, validation, and test datasets.
- Evaluated ICD and CPT prediction using Micro-F1, Macro-F1, and Precision@5, with Longformer achieving the strongest overall joint score of 0.5925 and the temporal BERT + LSTM model achieving a CPT Micro-F1 of 0.6096.
- Investigated temporal clinical reasoning across patient admissions, demonstrating how multi-note modeling can capture patient progression and provide more context-aware medical code predictions than independent note-level modeling.

COVID-19 Forecasting Web Application — **August 2025**
GitHub Repository | **Live Application**
Python, Prophet, Plotly

- Built and deployed an interactive COVID-19 analytics application, processing 161,568 records across 198 countries and 816 reporting dates.
- Developed data-cleaning and analysis workflows for confirmed cases, recoveries, deaths, country-level filtering, date-range selection, daily changes, and epidemiological ratios.
- Implemented country-level time-series forecasting for 1–5 year horizons using Prophet with weekly and yearly seasonality and 95% prediction intervals.
- Created interactive maps, country comparisons, KPI dashboards, trend charts, rankings, and downloadable reports using Plotly, Matplotlib, and Streamlit.